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III Semester BE Degree Examination

MODEL PAPER

OBJECT ORIENTED PROGRAMMING USING C++

Duration: 03 Hours **Max. Marks: 100**

Instructions:

- **All questions in PART-A is compulsory**
- **PART-B has internal choice**

PART – A				
Q.NO	**CO**	**CD**	**PI**	**MARKS**
1.	1	L3	1.4.1	10
	Write a C++ program to create a class Student with data members name, rollNo, and marks. Include member functions to read and display the details. Explain the structure of your C++ program.			
2.	2	L3	2.2.3	10
	Explain the concept of classes and objects in C++. Write a program to create a class Employee with data members empId, name, and salary. Include member functions to read and display employee details. Also explain the role of static data members and static member functions.			
3.	3	L2	1.4.1	10
	What is Inheritance? Explain with an example of inheriting the multiple base classes?			
4.	4	L2	2.2.3	4+6
	a) Explain the working of multiple catch blocks in C++ with syntax. **b)** Write a C++ program that throws an exception if a number entered by the user is negative and handles it appropriately.			
5.	5	L2	2.2.3	10
	Differentiate between function templates and class templates with suitable			

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	examples.

PART – B

Q.NO	CO	CD	PI	MARKS
6.	1	L2	2.2.3	10
	What is function overloading in C++? Explain its advantages. Write a C++ program to overload a function volume() to calculate the volume of a cube, a cylinder, and a rectangular box.			
	OR			
7.	1	L2	2.2.3	10
	Explain the basic concepts and benefits of Object-Oriented Programming (OOP). How does OOP overcome the limitations of Procedure-Oriented Programming (POP)? Illustrate your answer with a suitable example in C++.			
8.	2	L2	2.2.3	10
	What are constructors and destructors in C++? Explain different types of constructors with examples. Write a C++ program to demonstrate a parameterized constructor and a copy constructor.			
	OR			
9.	2	L3	4.2.1	10
	What is operator overloading in C++? Explain its rules. Write a C++ program to overload the binary + operator to add two complex numbers.Read and print the result by overloading >>,<< operators.			
10.	3	L3	4.2.1	10
	Explain how to pass arguments to base class constructors in multiple inheritance, with suitable example.			
	OR			
11.	3	L3	4.2.1	10
	What is a pure virtual function ?Explain with example the generic function using template.			
12.	4	L3	1.4.1	5+5
	Write the various functions used to open, close, read and write data in text files.			

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	Write a C++ program to create a file named **"PRODUCT"**, store product ID, product name, and price in it, and then read and display the stored details. Use constructor functions to open the file.			
OR				
13.	4	L3	4.2.1	10
	Write a function which throws a division by zero exception and catch it in catch block. Write a C++ program to demonstrate usage of try, catch and throw to handle exception.			

14.	5	L3	4.3.2	10
	Show the implementation of stack data structure using member function template.			
OR				
15.	5	L3	4.3.2	10
	What is a template?Write a C++ program to create a function template that finds the maximum of two numbers. Demonstrate the function using int, float, and double data types.			

Course Outcome (CO):	
CO–1	Understand object-oriented programming concepts and be able to design solutions for a given problem.
CO–2	Able to reuse code with extensible Class types, user defined operators and function overloading.
CO–3	Achieve code reusability and extensibility by means of Inheritance and Polymorphism.
CO–4	Incorporate file-handling functions and exceptions for a given application.
CO–5	Design C++ packages with templates for providing solutions to complex problems.

PI	
1.4.1	Apply theory and principles of computer science and engineering to solve an engineering problem

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2.1.2	Identify processes/modules/algorithms of a computer-based system and parameters to solve a problem
2.2.3	Identify existing solution/methods to solve the problem, including forming justified approximations and assumptions
3.2.2	Able to produce a variety of potential design solutions suited to meet functional requirements.
4.2.1	Design and develop appropriate procedures/methodologies based on the study objectives

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